WINDOWS SERVER ADMINISTRATION LAB • PROJECT 2

Trust Relationship Failure

Repairing a broken computer account secure channel without rejoining the domain

Problem → Investigation → Fix → Verification

PREPARED BY

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Environment	Lab date	Purpose
VMware lab, testuser.local (DC01 + WIN11-WS01)	October 18, 2026	Portfolio lab and troubleshooting record

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Report Overview

The ticket

> **ℹ USER REPORT**
> "I can't sign in to my PC. It says 'The trust relationship between this workstation and the primary domain failed.' It worked yesterday."

This is one of the most common tickets in a Windows domain. The user's password is fine, the network is fine, and other PCs work. The problem is between **this one computer and the domain controller**.

The key idea: computers have passwords too

Every computer joined to a domain has its own **computer account** in Active Directory (shown with a $ at the end, for example **WIN11-WS01$**) and its own secret **computer account password**. Before any domain user can sign in, the computer first proves itself to the domain controller with that password. This trusted connection is called the **secure channel**.

Windows changes the computer password **automatically about every 30 days**, and the computer and the domain controller update their copies at the same time. If the computer ends up holding an **older** password than Active Directory, for example after being restored from an old snapshot or backup, the domain controller rejects it and no domain user can sign in on that PC.

What I did in this project

1. Created a test user and proved the secure channel was healthy.
2. Recreated a real-world cause: took a VM snapshot, changed the computer password, then reverted the snapshot so the PC had an out-of-date password.
3. Investigated like a help desk technician: ruled out the network and DNS, tested the secure channel, and found evidence on both the PC (Event 3210) and the domain controller (Event 5722).
4. Repaired the trust in seconds with Test-ComputerSecureChannel -Repair, without removing the PC from the domain.
5. Verified the repair from both sides and confirmed the user could sign in.

Lab environment

Item	Value
Virtualization	VMware Workstation, NAT network 192.168.76.0/24
Domain	**testuser.local** (NetBIOS name **TESTUSER**)
Domain controller	**DC01**, Windows Server 2022, 192.168.76.138, AD DS + DNS
Workstation	**WIN11-WS01**, Windows 11 Pro, 192.168.76.137 (computer account **WIN11-WS01$**)

Item	Value
Test user	**IT User02** (`ituser02`), created for this project, never signed in on WIN11-WS01 before
Local admin on workstation	**Urvisha** (`.\Urvisha`)

> ✓ **RESULT IN ONE SENTENCE**
>
> WIN11-WS01 was restored to a snapshot taken before its computer password changed, so the domain controller rejected it (Event 5722). I confirmed the cause from both sides and repaired the secure channel with Test-ComputerSecureChannel -Repair, after which the domain user signed in normally.

Part A: Preparing the Lab

A1. Stop the previous project's task

▸ **Run on: WIN11-WS01 (PowerShell as Administrator, signed in as .\Urvisha)**

```
PS> Disable-ScheduledTask -TaskName "OneDrive Backup Sync"
```

What it does: Turns off the scheduled task from my previous lockout project so it cannot add unrelated failed sign-ins to the logs during this investigation.

A2. Create a fresh test user

In **Active Directory Users and Computers > Lab Users**, I created **IT User02** (logon name **ituser02**) and unticked "User must change password at next logon".

> ℹ **WHY A NEW USER?**
>
> IT User01 had already signed in on WIN11-WS01, so Windows keeps a **cached copy** of that sign-in and might let the user in even with a broken trust. IT User02 had never signed in on this PC, so there is no cached sign-in to fall back on and the real error is guaranteed to appear.

A3. Confirm the secure channel is healthy

▸ **Run on: WIN11-WS01**

```
PS> Test-ComputerSecureChannel -Verbose
```

What it does: Checks whether this computer and the domain controller still trust each other. **True** means healthy, **False** means broken. `-Verbose` adds a plain-English explanation.

> ⚠ **PROBLEM I HIT: A FALSE ALARM**
>
> The first time I ran it, Test-ComputerSecureChannel returned **False** ("the secure channel... is broken") before I had changed anything. Instead of repairing straight away, I checked with a second tool, `nltest /sc_verify:testuser.local`, which returned **NERR_Success**. Running Test-ComputerSecureChannel again then returned **True**.
>
> The connection had only gone stale (the VMs had been idle), and nltest re-established it. Confirming with two tools before acting avoided an unnecessary repair on a healthy computer.

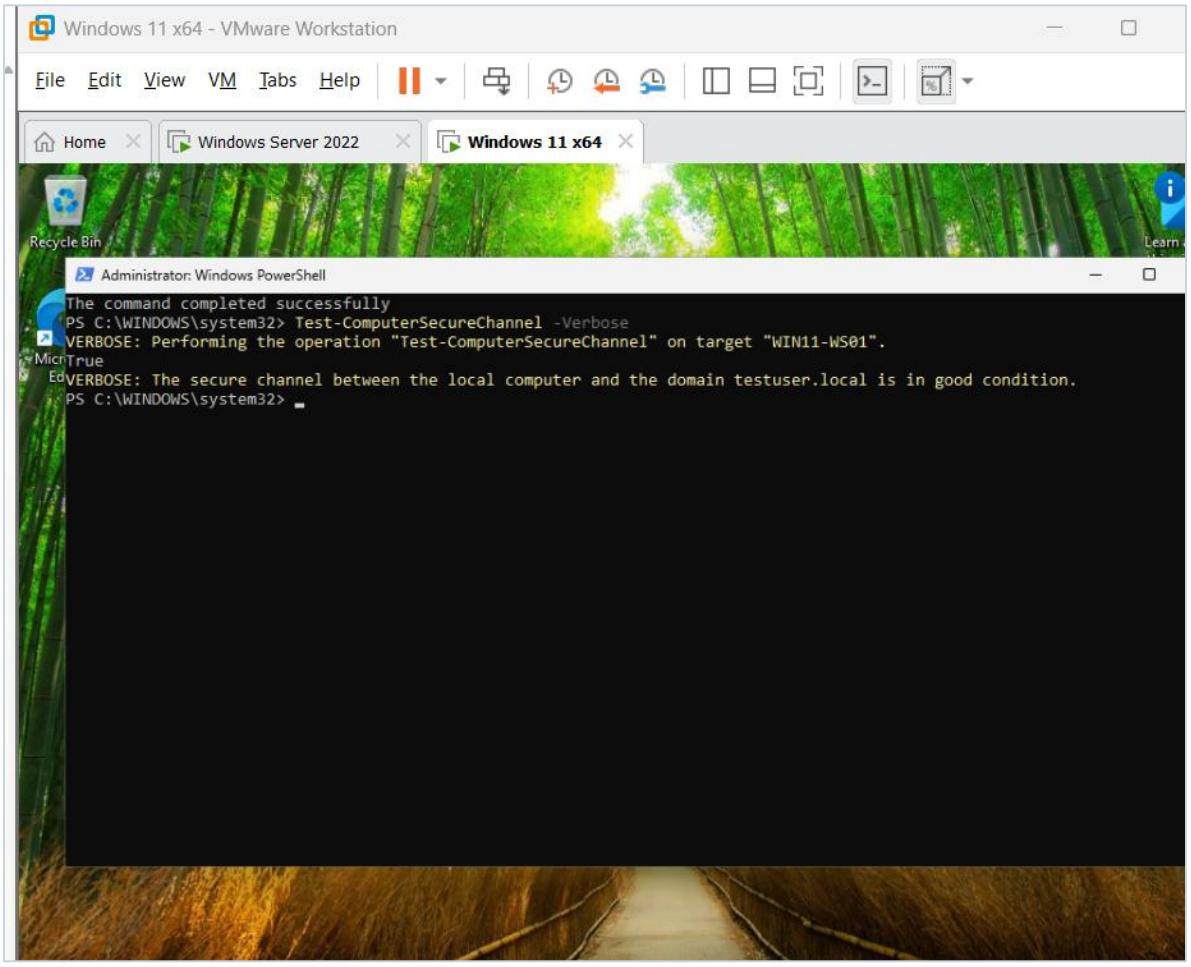


FIGURE 1 *Before: the secure channel between WIN11-WS01 and testuser.local is "in good condition" (True).*

A4. Record when the computer password was last set

▸ **Run on: DC01 (PowerShell as Administrator)**

```
PS> Get-ADComputer WIN11-WS01 -Properties PasswordLastSet | Select-Object Name,
PasswordLastSet
```

What it does: Shows the date and time WIN11-WS01's computer account password was last changed in Active Directory.

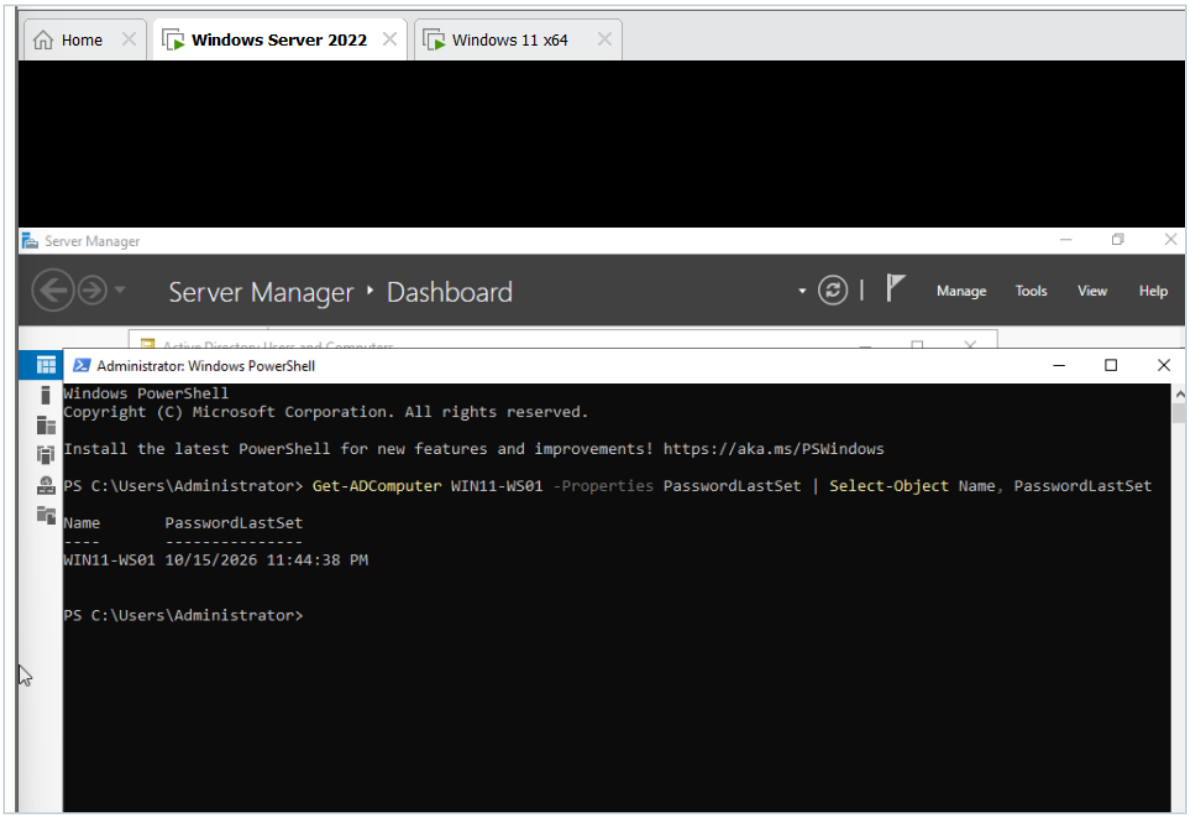


FIGURE 2 *The computer password was set on Oct 15, 2026 at 11:44:38 PM, the moment WIN11-WS01 joined the domain. Joining a domain creates the computer's first password.*

Part B: Planting the Problem

To recreate a realistic cause, I simulated a computer being **restored from an old backup** after Windows had changed its computer password.

B1. Take the "old backup"

In VMware: **VM > Snapshot > Take Snapshot**, named **Before-Trust-Break**. This snapshot holds the computer's current (soon to be old) password.

B2. Change the computer password

> ⚠ **PROBLEM I HIT: ACCESS WITH THE WRONG ACCOUNT**
>
> My first attempt, `Reset-ComputerMachinePassword -Server DC01.testuser.local`, failed with "The user name or password is incorrect". I was signed in as the **local** account Urvisha, which the domain does not know, and this command authenticates to the DC with the current user unless domain credentials are supplied with `-Credential`.
>
> I used `nltest /sc_change_pwd` instead, which uses the **computer's own account** (WIN11-WS01$), so no domain user is needed.

▸ **Run on: WIN11-WS01 (run twice)**

```
PS> nltest /sc_change_pwd:testuser.local
```

What it does: Makes the computer choose a new computer account password immediately and send it to the domain controller. Windows normally does this by itself about every 30 days; this command simply does it now. **NERR_Success** means the change worked.

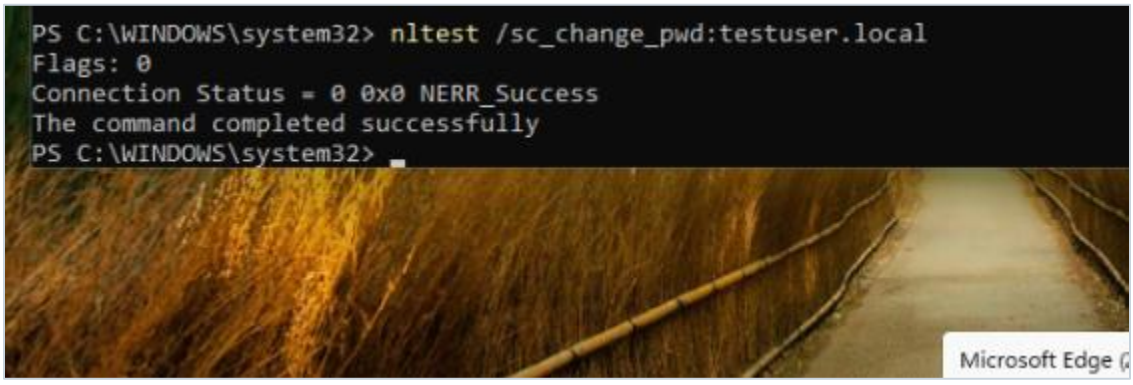


FIGURE 3 *The computer password change succeeded (NERR_Success).*

> ℹ **WHY RUN IT TWICE?**
>
> The domain controller still accepts a computer's **previous** password, much like Active Directory forgives a user's recent passwords. After two changes, the password saved in the snapshot is too old to be accepted.

▸ **Run on: DC01**

```
PS> Get-ADComputer WIN11-WS01 -Properties PasswordLastSet | Select-Object Name,
PasswordLastSet
```

What it does: Confirms the change reached Active Directory.

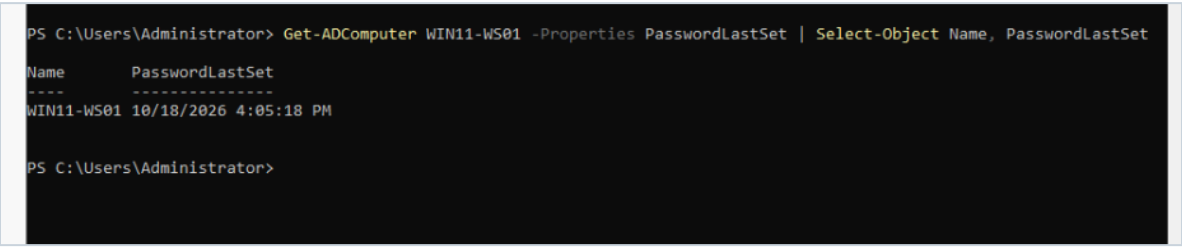


FIGURE 4 *Active Directory now has a new computer password, set at 4:05:18 PM.*

B3. "Restore from backup"

In VMware: **VM > Snapshot > Revert to Snapshot > Before-Trust-Break**, then I restarted WIN11-WS01. The PC was now back to its earlier state, holding the **Oct 15** computer password, while Active Directory held the **4:05 PM** one. The restart cleared any sign-in tickets still in memory, so the PC had to prove itself to the DC again.

> ℹ **HOW THIS HAPPENS FOR REAL**
>
> Nobody runs nltest /sc_change_pwd in normal use. In real life the change happens automatically every 30 days, and the problem appears when a PC or VM is later restored from a snapshot, image or backup taken before that automatic change. I used the command only to avoid waiting 30 days.

Part C: The Investigation

From here I worked as the help desk technician with only the ticket to go on.

C1. Confirm the symptom

At the sign-in screen I chose **Other user** and entered **TESTUSER\ituser02** with the correct password.

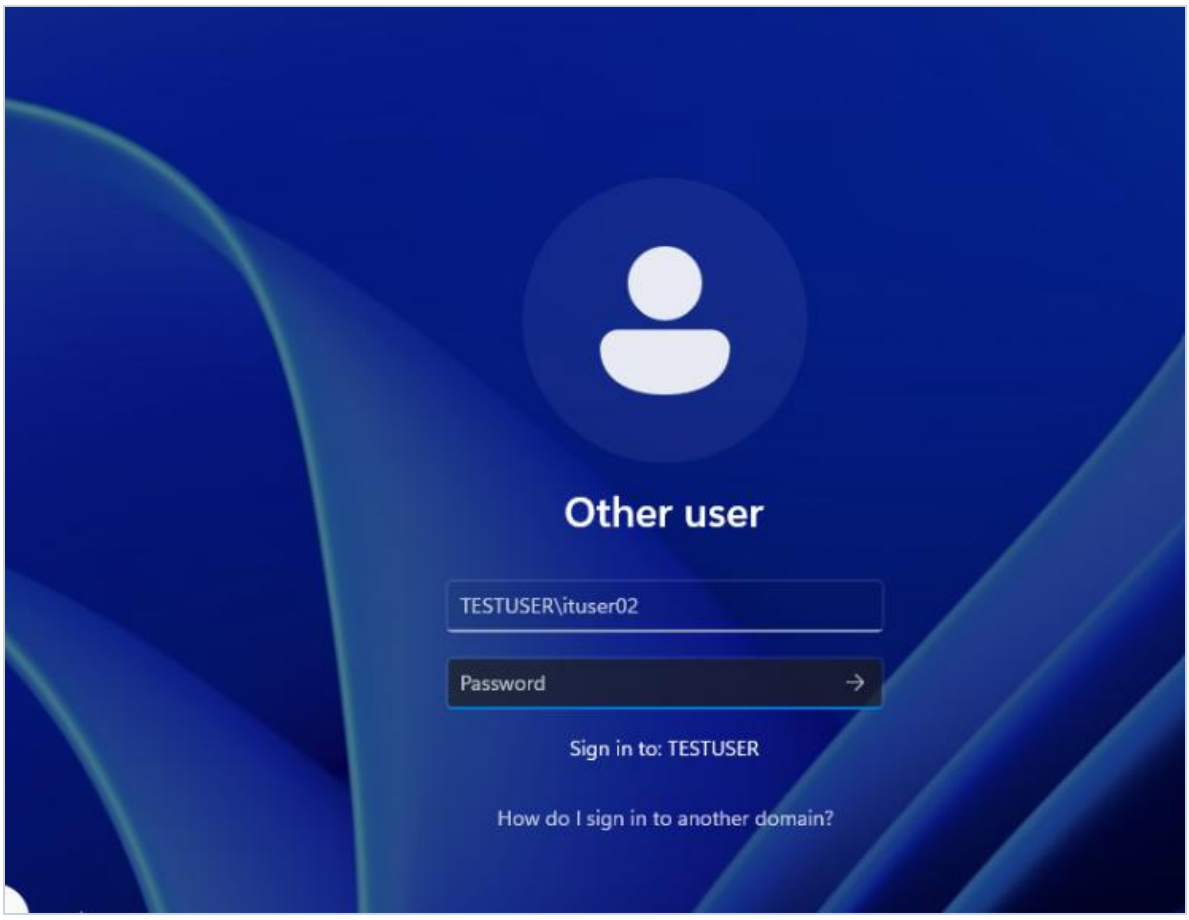


FIGURE 5 *Signing in to WIN11-WS01 as the domain user TESTUSER\ituser02.*

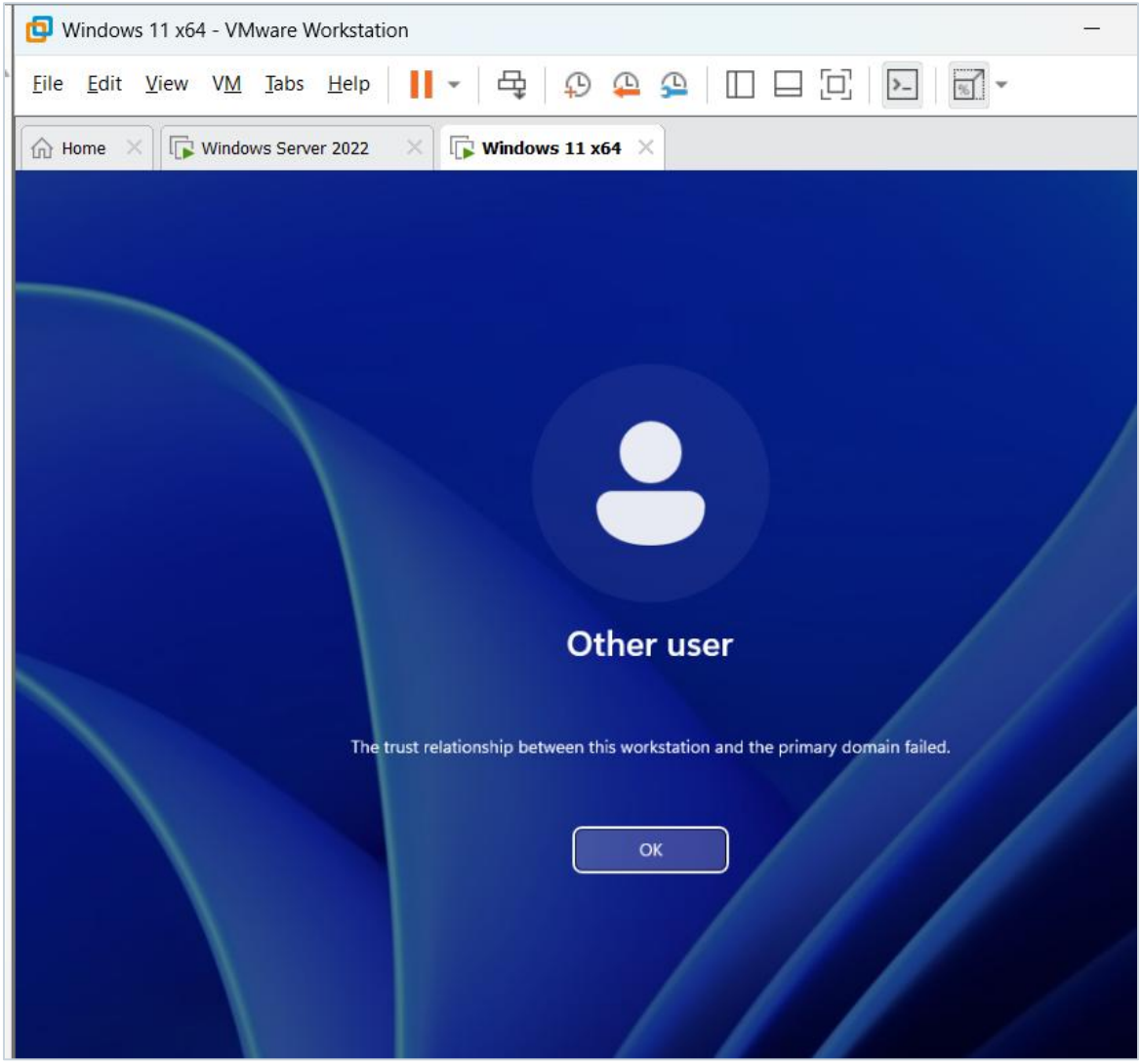


FIGURE 6 *The sign-in fails: "The trust relationship between this workstation and the primary domain failed."*

C2. Narrow it down

I signed in as the **local** administrator **.\Urvisha**, which worked. The computer itself was fine; the problem was only with **domain** sign-ins, so it was between the PC and the domain.

C3. Rule out the network and DNS

▸ **Run on: WIN11-WS01**

```
PS> ping 192.168.76.138
PS> nslookup testuser.local
```

What it does: `ping` checks the PC can reach DC01 over the network. `nslookup` checks DNS can find the domain. If either failed, the cause would be connectivity, not the trust.

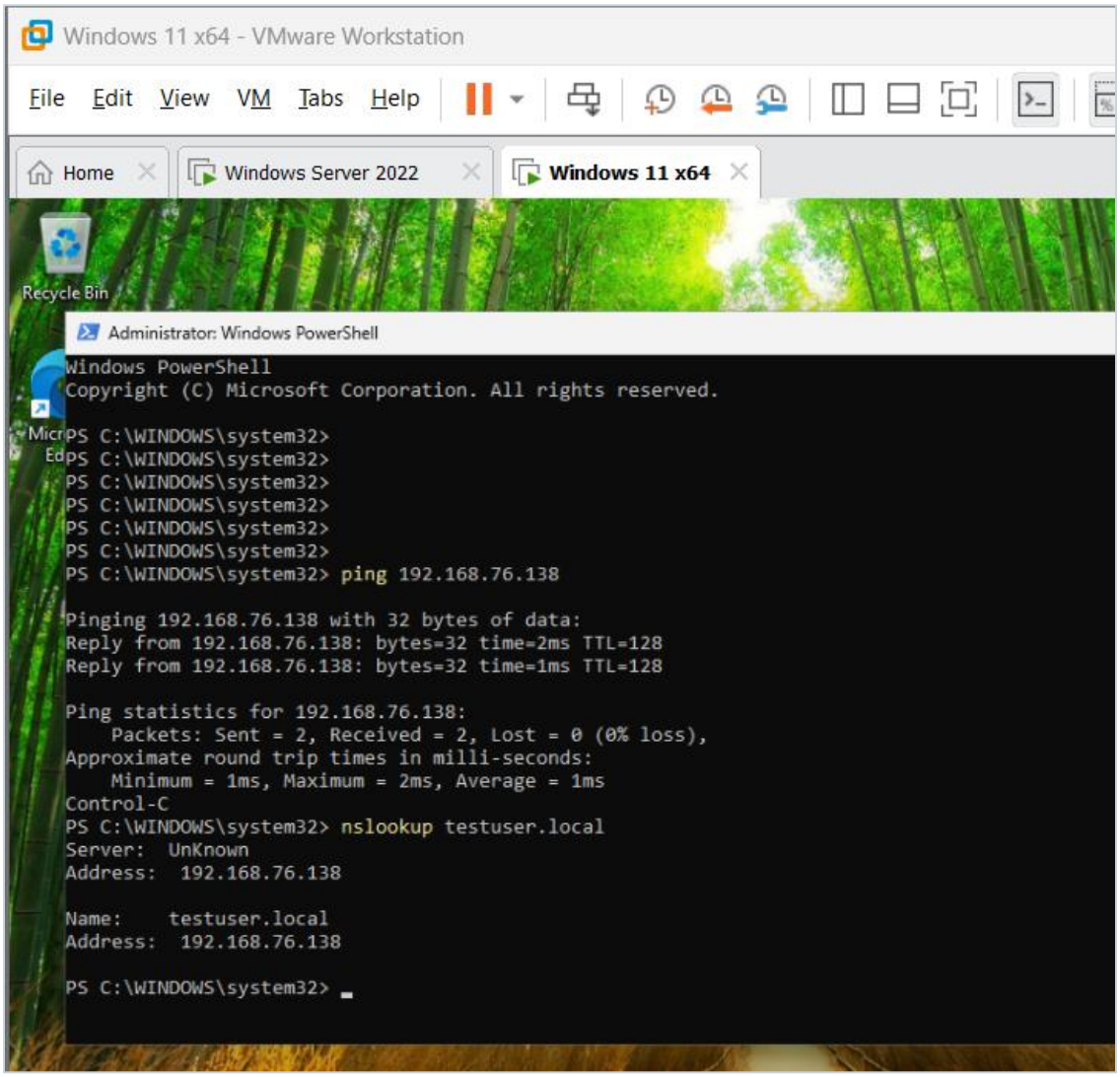


FIGURE 7 *DC01 replies to ping and DNS resolves testuser.local to 192.168.76.138. The network and DNS are working, so this is not a connectivity problem. (The "Server: UnKnown" line only means there is no reverse lookup zone, which is harmless.)*

C4. Test the secure channel

▸ **Run on: WIN11-WS01**

```
PS> Test-ComputerSecureChannel -Verbose
```

What it does: The same health check as in Part A.

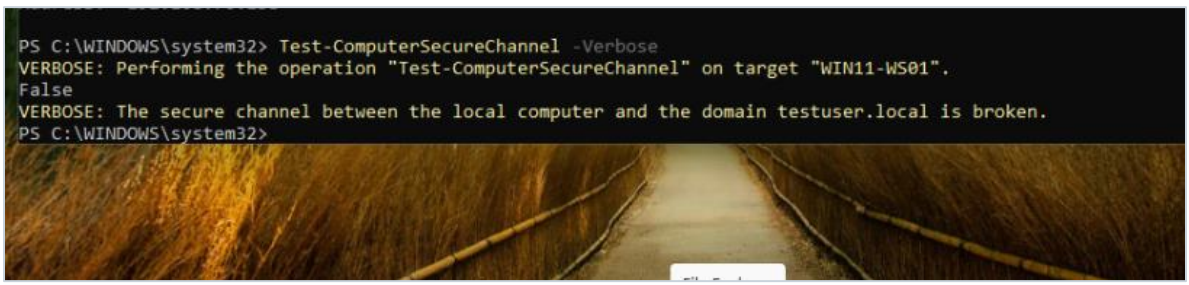


FIGURE 8 *Now the secure channel is broken (False), even though the network is fine.*

C5. Evidence on the PC: Event 3210

On WIN11-WS01: **Event Viewer > Windows Logs > System > Filter Current Log**, Event ID **3210** (source NETLOGON).

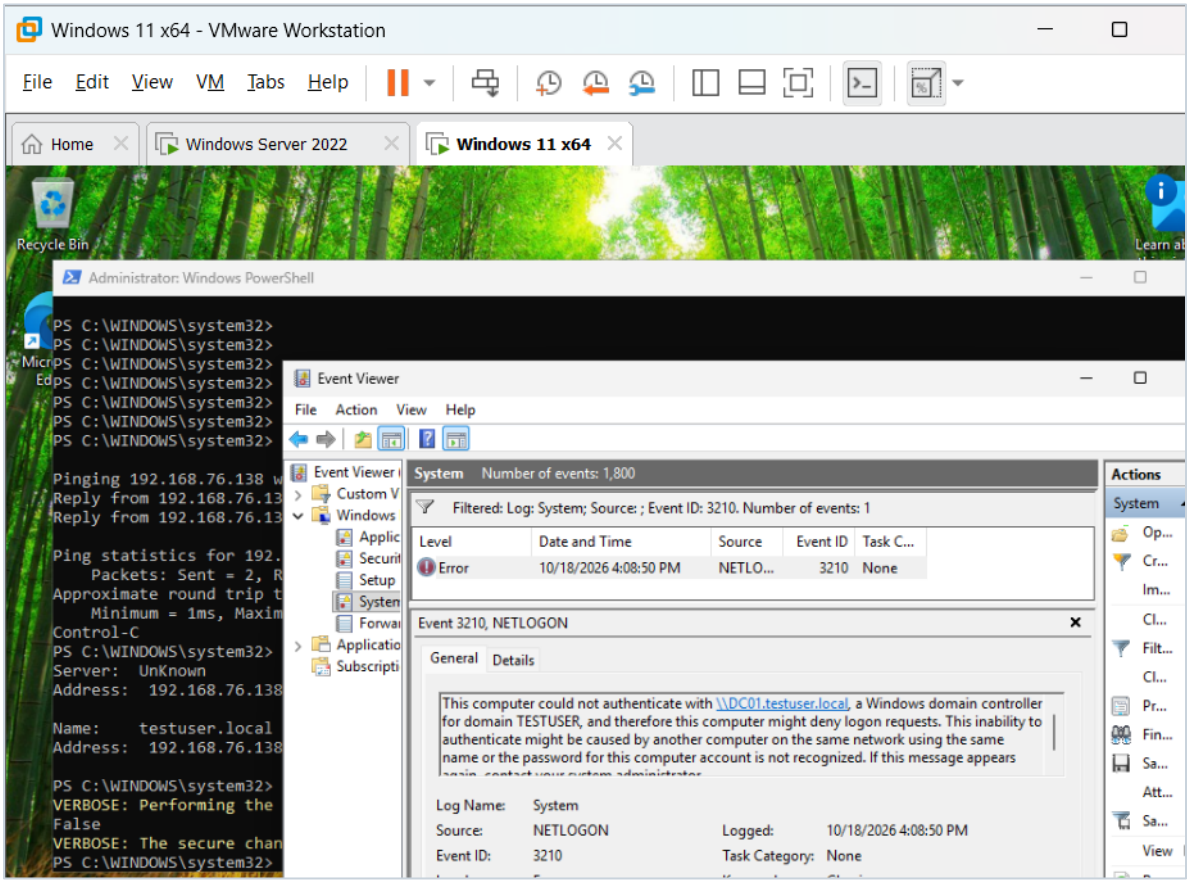


FIGURE 9 *Event 3210 at 4:08:50 PM: "This computer could not authenticate with \\DC01.testuser.local... the password for this computer account is not recognized."*

C6. Evidence on the domain controller: Event 5722

On DC01: **Event Viewer > Windows Logs > System > Filter Current Log**, Event ID **5722**. This event is recorded on the domain controller, not on the workstation; searching the workstation returns nothing.

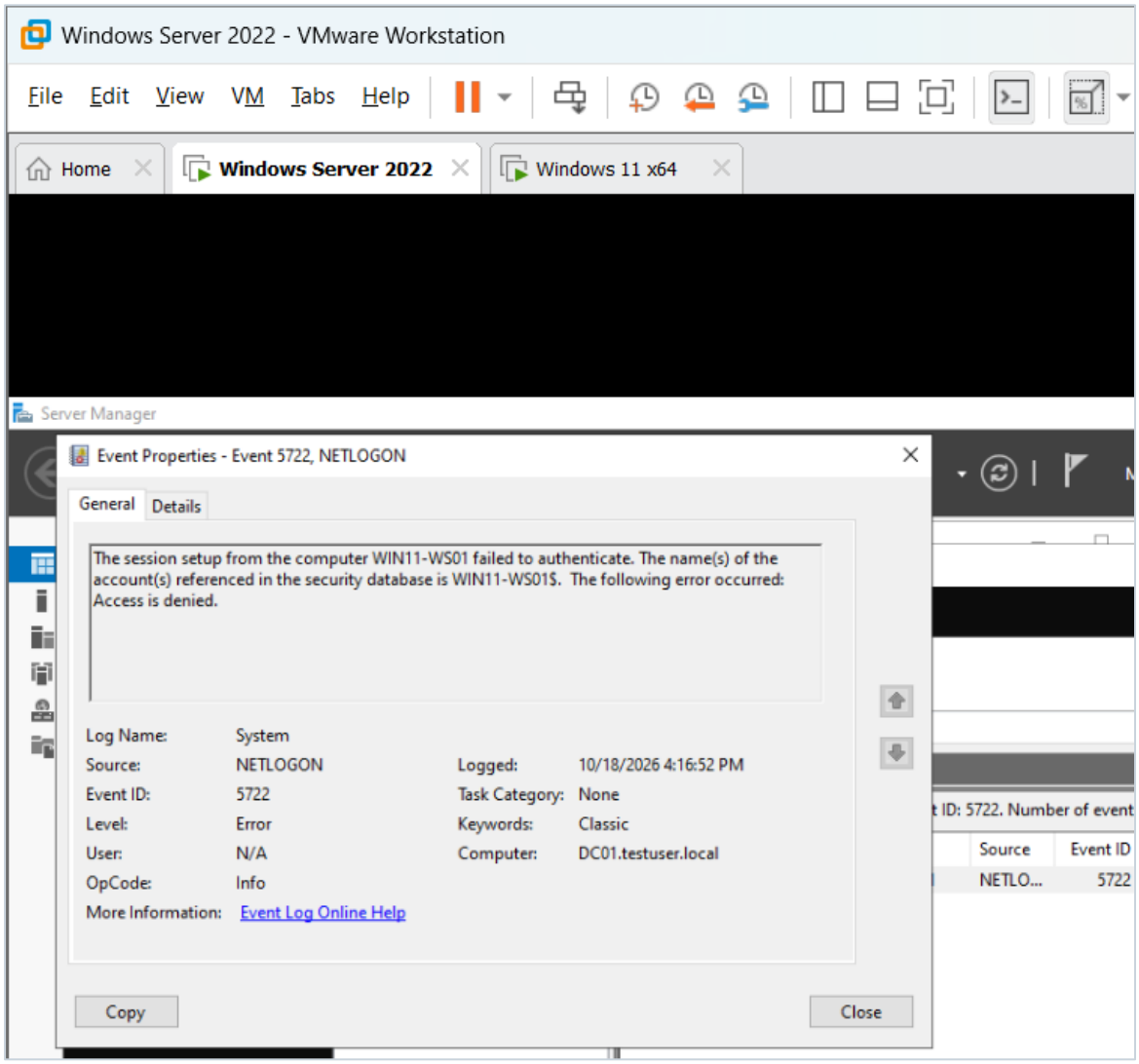


FIGURE 10 *Event 5722 on DC01 at 4:16:52 PM: "The session setup from the computer WIN11-WS01 failed to authenticate... WIN11-WS01$... Access is denied."*

Where	Event	What it says
WIN11-WS01	**3210** (NETLOGON)	"I could not authenticate with DC01; my computer password is not recognized."
DC01	**5722** (NETLOGON)	"WIN11-WS01$ tried to connect with the wrong password: Access is denied."

C7. Compare the passwords

▸ **Run on: DC01**

```
PS> Get-ADComputer WIN11-WS01 -Properties PasswordLastSet | Select-Object Name,
PasswordLastSet
```

What it does: Shows which computer password Active Directory expects.

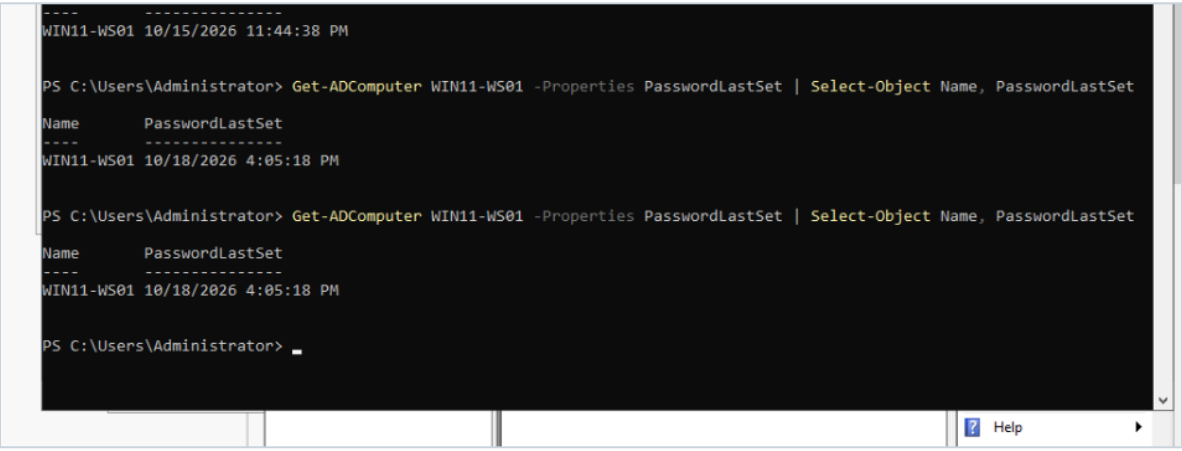


FIGURE 11 *Active Directory expects the 4:05:18 PM password, but the restored PC still holds the Oct 15 password from its snapshot.*

◉ CONFIRMED CAUSE

WIN11-WS01 was restored to a snapshot taken **before** its computer password changed. The PC presents the old password, the domain controller rejects it (**Event 5722**), the PC cannot authenticate (**Event 3210**), and domain users cannot sign in.

The network, DNS and the user's own password were all fine.

Part D: The Fix

▸ **Run on: WIN11-WS01 (PowerShell as Administrator, signed in as .\Urvisha)**

```
PS> Test-ComputerSecureChannel -Repair -Credential TESTUSER\Administrator
```

What it does: Creates a **new** computer password and saves it on **both** the PC and the domain controller at the same time, so they match again. `-Credential` supplies a domain account that is allowed to reset the computer account, because I was signed in as a local user. Windows prompts for that account's password.

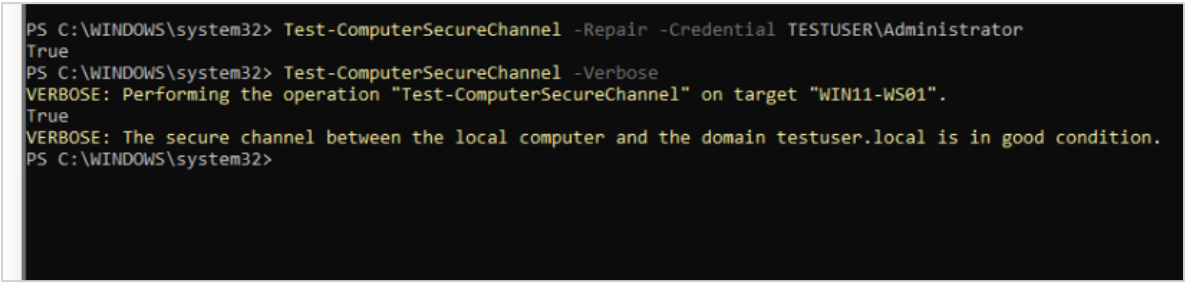


FIGURE 12 *The repair returns True, and the follow-up check reports the secure channel is "in good condition".*

Why this fix instead of rejoining the domain

Method	How	Downside
Test-ComputerSecureChannel -Repair (used)	One PowerShell command on the PC	Needs a local admin sign-in and domain credentials
Reset-ComputerMachinePassword -Credential	Resets the computer password with domain credentials	Same result; the check must be run separately
Remove from domain and rejoin	Join a workgroup, restart, join the domain again, restart	Two restarts, longer outage for the user, and the computer account can lose settings if deleted

> **ℹ CHANGE MADE**
> Repaired the secure channel with Test-ComputerSecureChannel -Repair. No restart, no domain rejoin and no change to any user account.

Part E: Final Verification

▸ **Run on: WIN11-WS01**

```
PS> nltest /sc_verify:testuser.local
```

What it does: Verifies the secure channel with the older tool, as a second, independent check. **Trusted DC Connection Status** shows the PC can reach the DC; **Trust Verification Status** shows the DC accepts the PC's password.

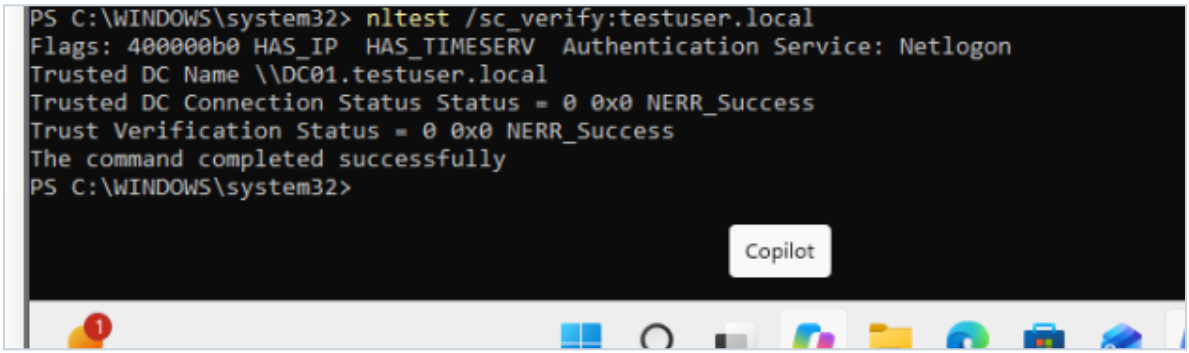


FIGURE 13 *Both statuses report NERR_Success: the trust with DC01.testuser.local is healthy.*

▸ **Run on: DC01**

```
PS> Get-ADComputer WIN11-WS01 -Properties PasswordLastSet | Select-Object Name,
PasswordLastSet
```

What it does: Confirms Active Directory received the new password from the repair.

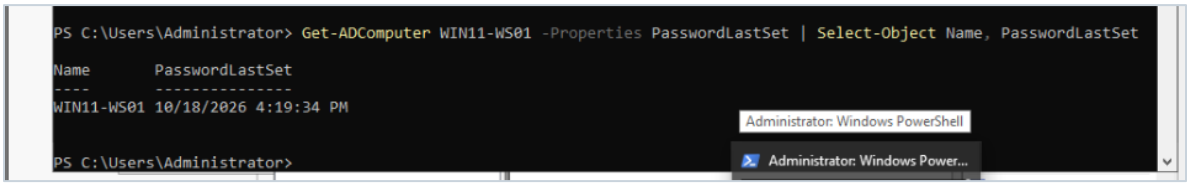


FIGURE 14 *PasswordLastSet is now 4:19:34 PM, the time of the repair. The PC and Active Directory hold the same new password.*

Finally I signed out and signed in as **TESTUSER\ituser02**.

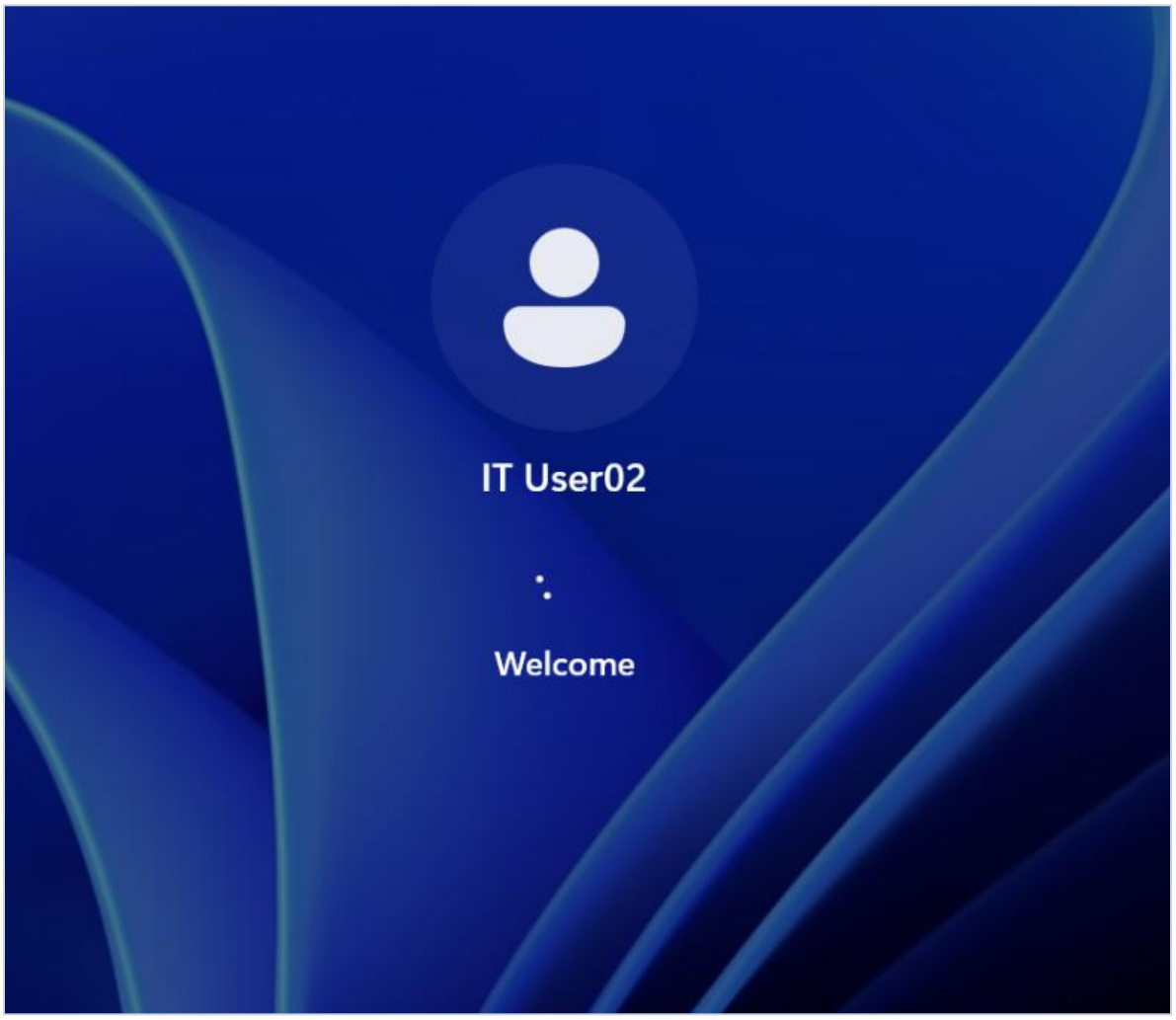


FIGURE 15 *IT User02 signs in successfully ("Welcome").*

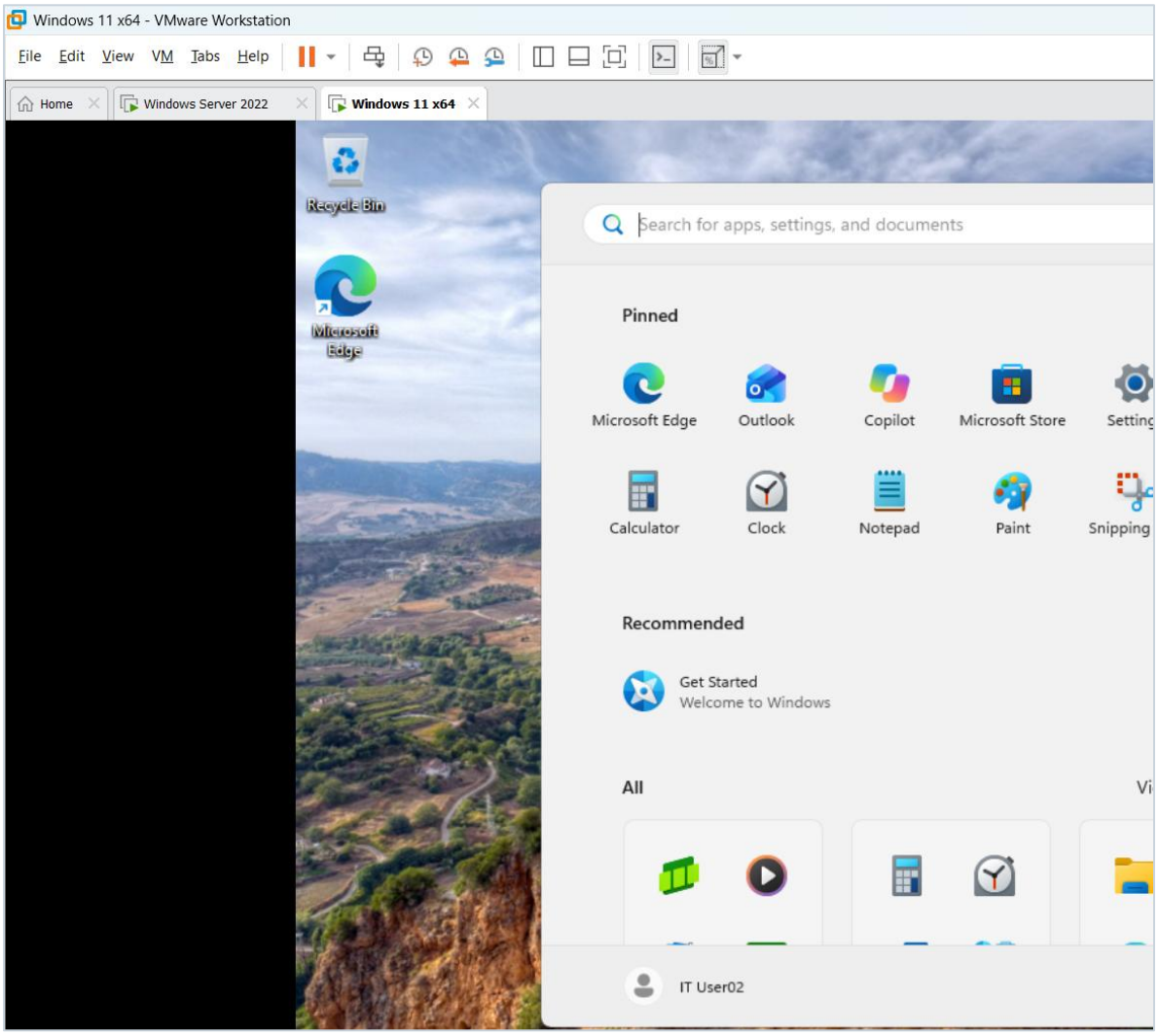


FIGURE 16 *The IT User02 desktop on WIN11-WS01, with the Start menu showing IT User02 as the signed-in user. The domain sign-in works again.*

✓ FINAL RESULT

The secure channel is healthy (Test-ComputerSecureChannel True, nltest NERR_Success), Active Directory shows the new computer password, and the domain user signs in and reaches the desktop normally.

Incident Timeline

Date / time	Event
Oct 15, 11:44:38 PM	WIN11-WS01 joins the domain; first computer password created
Oct 18, before 4:05 PM	Secure channel confirmed healthy; snapshot "Before-Trust-Break" taken
Oct 18, 4:05:18 PM	Computer password changed (nltest /sc_change_pwd)
Oct 18, ~4:06 PM	Snapshot reverted and PC restarted: PC holds the old password
Oct 18, 4:08:50 PM	**Event 3210** on WIN11-WS01: cannot authenticate with DC01
Oct 18, ~4:10 PM	IT User02 sign-in fails: "trust relationship ... failed"
Oct 18, 4:16:52 PM	**Event 5722** on DC01: WIN11-WS01$ rejected, Access is denied
Oct 18, 4:19:34 PM	**Repair**: new computer password set on both sides
Oct 18, after 4:19 PM	nltest NERR_Success; IT User02 signs in

Computer password at each stage

Stage	Password on WIN11-WS01	Password in Active Directory	Trust
Before the lab	Oct 15	Oct 15	Healthy
After nltest /sc_change_pwd	4:05 PM	4:05 PM	Healthy
After reverting the snapshot	**Oct 15**	**4:05 PM**	**Broken**
After -Repair	4:19 PM	4:19 PM	Healthy

Command Reference

Command	Where	Purpose
`Disable-ScheduledTask -TaskName "..."`	WIN11-WS01	Turn off a scheduled task
`Test-ComputerSecureChannel -Verbose`	WIN11-WS01	Check whether the PC and the domain still trust each other
`Test-ComputerSecureChannel -Repair -Credential DOMAIN\Admin`	WIN11-WS01	Reset the computer password on both sides to fix the trust
`nltest /sc_verify:testuser.local`	WIN11-WS01	Verify (and re-establish) the secure channel
`nltest /sc_change_pwd:testuser.local`	WIN11-WS01	Change the computer password now (normally automatic every 30 days)
`Reset-ComputerMachinePassword -Server DC01... -Credential ...`	WIN11-WS01	Change the computer password with domain credentials

Command	Where	Purpose
`Get-ADComputer WIN11-WS01 -Properties PasswordLastSet`	DC01	Show when the computer password was last set in Active Directory
`ping 192.168.76.138`	WIN11-WS01	Test network reachability to the DC
`nslookup testuser.local`	WIN11-WS01	Test that DNS can find the domain

Other useful nltest commands

Command	What it does
`nltest /sc_query:testuser.local`	Show the current secure channel status without re-checking
`nltest /sc_reset:testuser.local`	Drop and rebuild the secure channel with a DC
`nltest /dsgetdc:testuser.local`	Find a domain controller (useful for DNS problems)
`nltest /dclist:testuser.local`	List the domain controllers

Graphical tools used

Tool	How to open	Used for
Event Viewer	Win + R > eventvwr.msc	Events 3210 (PC) and 5722 (DC) in the System log
Active Directory Users and Computers	Server Manager > Tools	Creating IT User02
VMware Snapshot Manager	VM > Snapshot	Taking and reverting the snapshot

How This Happens in Real Companies

Cause	What happens
PC or VM restored from an old **snapshot, image or backup**	The restored copy has a computer password older than Active Directory (the cause in this lab)
An admin **resets the computer account** in ADUC	Active Directory gets a new password the PC does not know
The computer account is **deleted and recreated**	A new account with a different password and identity
Two PCs with the same name join the domain	The second join overwrites the first PC's password
A PC is **cloned** without sysprep	Copies share or carry an outdated computer identity

> **ℹ A COMMON MYTH**
>
> A PC that has been switched off for a long time does **not** lose its trust by itself. The PC starts the 30-day password change, so it simply changes its password when it comes back online.

Comparison with a user account lockout

	User account lockout	Trust relationship failure
Whose password?	A user's	A computer's (WIN11-WS01$)
What goes wrong	Something keeps using an old user password	The PC holds an old computer password
Key events	4740, 4771, 4625	3210 (PC), 5722 (DC)
Fix	Stop the source, then unlock the account	Test-ComputerSecureChannel -Repair

What I Learned

- **Computers sign in to the domain too.** Each has a computer account (NAME$) and a password that Windows changes automatically about every 30 days.
- **"Trust relationship failed" is not a user problem.** The user's password and the network can be perfectly fine.
- **Narrow it down first:** a local account sign-in, ping and nslookup separate a trust problem from a PC or network problem.
- **Look at both sides:** Event 3210 on the PC and Event 5722 on the DC tell the same story from each end.
- **Confirm with two tools.** Test-ComputerSecureChannel once reported a stale connection as broken; nltest showed it was healthy. Acting on one result would have meant an unnecessary repair.

- **The DC accepts the previous computer password**, so one password change is not enough to break the trust.
- **The right account matters:** Reset-ComputerMachinePassword failed as a local user, while nltest /sc_change_pwd uses the computer's own account.
- **Repair, don't rejoin:** Test-ComputerSecureChannel -Repair fixes the trust in seconds without restarts.

Summary and Recommendations

A domain user could not sign in because WIN11-WS01 had been restored to a snapshot taken before its computer password changed. After ruling out the network and DNS, I confirmed the broken secure channel and found matching evidence on the PC (Event 3210) and the domain controller (Event 5722). I repaired the secure channel with Test-ComputerSecureChannel -Repair and verified the fix with nltest, Active Directory and a successful user sign-in.

Recommendations

- **Be careful when reverting snapshots of domain-joined machines.** After reverting a workstation or member server, check the secure channel straight away and repair it if needed.
- **Never restore a domain controller from a VM snapshot** without a supported recovery process; it can cause much bigger Active Directory problems (USN rollback).
- **Use -Repair before rejoining.** It is faster for the user and keeps the computer account and its group memberships intact.
- **Avoid duplicate computer names and always sysprep images** before deploying them.
- **Monitor Event 5722 on domain controllers** to spot broken computer trusts before users call the help desk.
- **Keep a known local administrator account** (with a strong, managed password, for example via Windows LAPS) so technicians can sign in to repair a PC that has lost its trust.